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IGNITION TRANSIENT ANALYSIS OF A SOLID ROCKET MOTOR USING A ONE DIMENSIONAL TWO FLUID MODEL

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Abstract

A one dimensional two fluid numerical model has been used to study the ignition transient stage of a Space Shuttle solid rocket motor. During the ignition phase of a solid rocket motor a pressure transient is induced by complex transport processes involving the igniter gas heat transfer to the propellant, chemical reactions at the propellant surface, and the interaction of the fluid with the attached rocket nozzle. One dimensional models used in the past neglected the aluminum oxide particles which are present in the combustion gases. The current model uses the IPSA (Inter-Phase-Slip-Algorithm) to solve the transient compressible flow equations for the rocket chamber and attached nozzle. Numerical results for head end pressure changes and overall thrust are compared with both measurement data and predictions of a one dimensional one fluid model.

Introduction

The ability to predict and control the complex transport processes in the rocket chamber is desired for the efficient design of rocket motors. Engineering design parameters include: overall transient time of the motor, reduction of motor to motor variation, over pressurization, avoiding misfires and hangfires, effects of design modifications, and variation of both total and dynamic thrust.

The ignition transient stage of a Space Shuttle solid rocket motor is the focus of this study. As the pyrogen gas from the igniter heats the surface of a solid propellant, the temperature of the fuel increases until the propellant surface reaches its auto-ignition temperature. Energy released through the combustion of the fuel initiates a strong convection in the chamber which in turn spreads the flame toward the aft-end of the rocket chamber. Following the completion of flame spreading, the gas pressure in the chamber increases rapidly due to the energy

and mass released from the burning solid propellant. Ignition (or starting) transient refers to the short period of motor operation from the start of igniter gas flow to the end of rapid chamber pressure build up.

Although the true flow in the rocket is complex, one dimensional gas dynamic models have been successfully used to analyze the ignition transient of high velocity solid rocket motors^{1,2,3,4}. Peretz et. al.¹ performed both experimental and analytical modeling of solid rocket motors employing a predictor-corrector numerical method. Caveny and Kuo² extended these techniques to the Space Shuttle SRM. Caveny³ further extended this model and compared numerical predictions with Shuttle SRM test data. Total thrust build-up predicted by Caveny and Kuo's model agrees qualitatively with the experimental data. However, predicted transient pressure in the head-end of the motor shows extreme departure. Han⁴ developed a numerical model based on the finite volume approach to solve the one dimensional transient flow equations. This model offers an improvement over the Caveny and Kuo model in that it includes both the converging and diverging nozzle sections, and it accounts for erosive burning effects.

Although one dimensional models are quite limited, they offer simplicity of formulation and can be used effectively for parametric studies. To make the one dimensional model useful, accurate correlation data for the heat transfer and erosive burning rate must be provided. Usually, measurement data obtained under quasi-steady state conditions are used for these correlations in one-dimensional approaches.

Another aspect in the modeling of the flow in the rocket chamber is the presence of aluminum oxide particles in the combustion gases. The aluminum particles' primary purpose is to increase the solid rocket's specific impulse and propellant density. The presence of the particles has an effect on the flow dynamics in the motor, especially in the nozzle region. The addition of the two fluid effects to the numerical model should increase the accuracy of the simulations.

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A one dimensional, transient, compressible flow numerical model based on the SIMPLE algorithm was developed to analyze the flow and thermal behavior in solid rocket motors^{4,5}. In the two fluid model, the conservation of mass, momentum and energy equations for the aluminum oxide particles are added to the numerical formulation and resulting system of equations is solved by following the IPSA⁶ approach. The objective of the present study is to apply the model to study the effects of aluminum particles commonly used in solid propellants. The effects of particle density and particle loading on the ignition transients of a Space Shuttle SRM are investigated.

Analysis

Geometry

The Space Shuttle solid rocket motor is taken as the reference geometry for this investigation. The Shuttle SRM has a large length to diameter ratio and a small port to nozzle throat area ratio. The rocket motor geometry is shown in Figure 1.

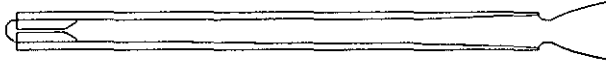


Figure 1. Computational geometry

The star shaped head-end segment is modeled by defining a burning perimeter ratio, α . This is given by

$$\alpha = \frac{\text{Actual burning surface perimeter}}{\text{Circumference of the cylinder}} \quad (1)$$

where $0 \leq \alpha \leq 6.7$.

Assumptions

The combustion products are assumed to be a mixture of an ideal gas and aluminum oxide particles. Chemical reaction occurs at an imaginary thin layer at the surface of the solid propellant when the fuel has reached its auto-ignition temperature. Transport coefficients are assumed to remain constant, and radiation effects are not accounted for. Flow and temperature fields are represented by one dimensional equations. Interactions between the combustion products and the solid surface are assumed to be represented by empirical momentum and energy transfer correlations.

Governing Equations

Conservation of mass is

$$\frac{\partial}{\partial t}(r_i \rho_i A) + \frac{\partial}{\partial x}(r_i \rho_i u_i A) = (w_i \rho_{PR} - r_i \rho_i) r b + \dot{m}_{ig} \quad (2)$$

where the subscript i indicates the i component (for this study component 1 is the gaseous products and component 2 is aluminum oxide particles), A is the cross-sectional area of the port, r_i is the volume fraction, ρ_i is the material density, u_i is the velocity, w_i is the mass fraction in the propellant, ρ_{PR} is the density of the propellant, b is the perimeter of the burning cross-section, $\dot{m}_{ig}$ is the igniter gas flow rate. Burning rate of the propellant is given by²

$$r = r_{ref} \left[\frac{P}{P_{ref}} \right]^n + \alpha_e G^{0.8} D_h^{0.2} \exp(-\beta_e r \rho_{PR} / G) \quad (3)$$

where $G = |\rho_i u_i|$, D_h is the hydraulic diameter, α_e and β_e are constants. The first term of the right hand side of equation 3 is the standard burning rate and the second term is the erosive burning rate. Conservation of mass also requires that

$$\sum r_i = 1.0 \quad ; \quad \sum w_i = 1.0 \quad (4)$$

Conservation of linear momentum is

$$\frac{\partial}{\partial t}(r_i \rho_i u_i A) + \frac{\partial}{\partial x} \left[A(r_i \rho_i u_i^2 - r_i \mu_i \frac{\partial u_i}{\partial x}) \right] = -r_i A \frac{\partial P}{\partial x} - \tau_w P_w - r_i \rho_i r b u_i + A F_{ij} \quad (5)$$

The thermodynamic pressure of the system is due to the gas phase only and is given by

$$P = \rho_i R_i T_i f(\rho_i) \quad (6)$$

where $f(\rho_i)$ is a correction needed for high pressure gas⁷. The wall shear stress is given by

$$\tau_{w_i} = \frac{1}{8} f \rho_i u_i^2 \quad (7)$$

where f is the friction factor given by²

$$f = \frac{0.449 (D_h / x)^{0.1}}{\left\{ \ln \left[\frac{\epsilon_s / D_h}{3.7} + \frac{1.25 R W^{0.5} T_{af}^{1.65} (D_h / x)^{0.05}}{10^6 P u_1 D_h f^{0.5}} \right] \right\}^2} \quad (8)$$

Shear stress due to the particle phase is assumed to be negligible.

The inter-phase friction force is given by⁸

$$F_{21} = \left[\frac{18 r_2}{d_2^2} \right] \mu_1 (u_2 - u_1) \left[1 + \frac{Re^{2/3}}{6} \right] \quad (9)$$

where d_2 is the particle diameter and the Reynolds number is defined by

$$Re = \frac{\rho_1 |u_2 - u_1| d_2}{\mu_1} \quad (10)$$

Energy conservation is, in terms of internal energy,

$$\begin{aligned} \frac{\partial}{\partial t} (r_i \rho_i e_i A) + \frac{\partial}{\partial x} \left[A (r_i \rho_i e_i u_i - r_i \frac{k_i}{C_v} \frac{\partial e_i}{\partial x}) \right] = \\ -r_i P \frac{\partial (u_i A)}{\partial x} + u_i \tau_w P_w + \Phi_i - q_1 P_w + A Q_{ij} \\ + A F_{ij} u_{ij} + w_i \rho_{PR} r b h_{\beta} + \dot{m}_{ig} h_{ig} \quad , \end{aligned} \quad (11)$$

where k_i is the thermal conductivity, h_{β} is the enthalpy of formation, h_{ig} is the enthalpy of the igniter gas, and Φ_i is the dissipation. The inter-phase heat transfer is calculated by⁸

$$Q_{21} = \left[\frac{6 k_1 r_2}{d_2^2} \right] (T_2 - T_1) (0.58 Re^{0.7} Pr^{0.3}) \quad (12)$$

where

$$Pr = \frac{\mu_1 C_{P_1}}{k_1} \quad (13)$$

Heat transfer from the gas to the solid propellant, when the propellant is below the auto-ignition temperature, is determined by

$$q_1 = h_c (T_1 - T_{surface}) \quad (14)$$

where h_c is an experimentally measured convective heat transfer coefficient given by²

$$h_c = 1.56 \times 10^{-3} C_{P,1} Pr^{-0.6} T_{af}^{-0.67} \left[\frac{P u_1}{R} \right]^{0.8} (x D_h)^{-0.1} \quad (15)$$

For the condensed phase, one dimensional conduction through the solid propellant is governed by

$$\rho_{PR} C_{PR} \frac{\partial T_{PR}}{\partial t} = \frac{1}{r} \frac{\partial}{\partial r} \left[r k_{PR} \frac{\partial T_{PR}}{\partial r} \right] \quad (16)$$

Initial and Boundary Conditions

Initially stagnant atmospheric air fills the rocket chamber and nozzle. The transient process begins with the introduction of igniter gas into the chamber at the head-end section. The pyrogen igniter mass flow rate is shown in Figure 2. Heat transfer from the gas to the solid propellant raises the temperature of the propellant resulting in auto-ignition. The ignition temperature of the propellant is assumed to be 850 K. The adiabatic flame temperatures of the igniter gas and solid propellant are taken as 2450 K and 3361 K, respectively. Mass and energy released from the burning fuel rapidly increase the momentum and energy of the flow in the system. Open boundary conditions are maintained at the exit nozzle and solid wall conditions are used at the head end. For the condensed phase, convective boundary conditions are assumed on the gas side and adiabatic conditions at the other side.

Numerical Method

IPSA (Inter-Phase-Slip-Algorithm) originated by Spalding⁶ is used to approximate the solution of the governing equations. IPSA is an extension of the SIMPLE method and consequently follows similar computational procedures. Equation 2 is solved for the volume fraction of the particle phase and then equation 4 is used to find the volume fraction of the gas phase. With the known volume fractions and an assumed pressure field, the momentum equations are solved for the velocities of each fluid. Corrections of density, velocity, and pressure are made by solving the pressure correction equation. Then the energy equation is solved to update the temperature. These procedures are repeated until convergence is satisfied within a given time step.

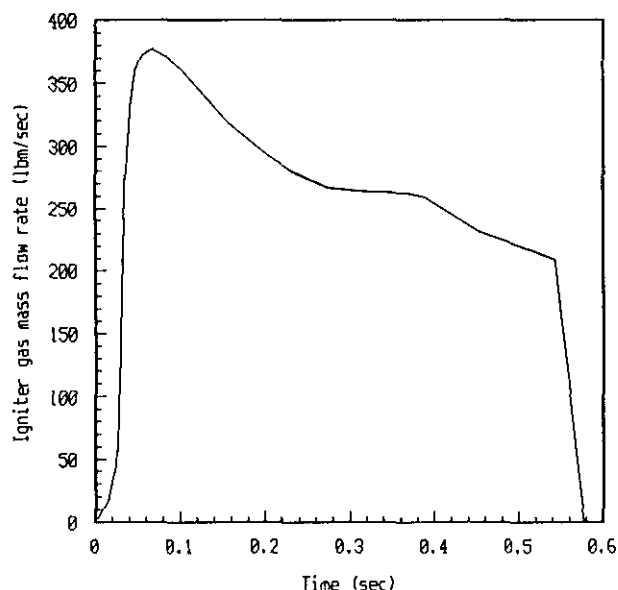


Figure 2. Igniter gas mass flow rate

Results

Numerical results are obtained from the two fluid model for two particle densities (1500 kg/m^3 and 3000 kg/m^3) and two particle loadings (10% and 17%). It should be noted that the flame temperatures used are for 10% aluminum particles in the propellant, and that the flame temperature is not adjusted for the 17% loading case. Therefore, the simulation results do not reflect the increase in thrust due to the additional aluminum particles. Although this does not provide accurate results, it does provide insight into the effect of aluminum oxide loading in the combustion gases on the overall rocket performance. The aluminum oxide particle diameter is assumed to be $10 \mu\text{m}$.

Figure 3 shows the influence of the number of control volumes on the thrust predictions. The grid size has a small effect on the results; however, the denser grid seems to provide better results overall and is used in this study. The program was run on a VAX 8800 and the required CPU times are given in Table 1.

Table 1. CPU time for various grid densities

Number of control volumes	CPU time (min)
61	21
76	29
94	40

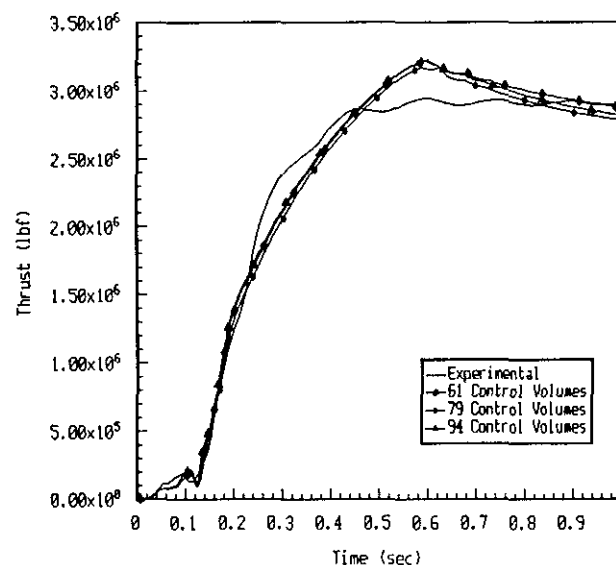


Figure 3. Influence of control volume size

Thrust predictions for $\rho = 1500$ and 3000 kg/m^3 and a particle loading of 10% are shown in Figure 4. Although the ignition transient ends at 0.6 seconds, the simulation is continued to 1.0 sec to show that the overshoot is corrected. The model provides both qualitative and quantitative agreement with the experimental thrust data for each density. As would be expected, the denser particles lower the thrust results. The effect of the same parameters on the head-end pressure gradient is given in Figure 5. The model provides qualitative agreement with the test data, and particle density seems to have very little effect on the predicted pressure gradient.

The influence of particle loading on the thrust predictions of the model are given in Figure 6. The low thrust predicted by the 17% loading is due in a large part to the flame temperature not being corrected as discussed earlier. Better results could be obtained by the proper representation of the flame temperature as a function of aluminum particles present. The results do show that the amount of particles present in the combustion gases can have a large effect on rocket performance. The predictions for the head-end pressure gradient for the same particle loading parameters is shown in Figure 7. The particle loading does not have as dramatic an effect of the pressure gradient as it did on the thrust. However, the amount of particles does show an effect that can not be neglected.

Figure 5. Head-end pressure gradient for $p = 1500$ and 3000 kg/m^3 and 10% aluminum oxide particles

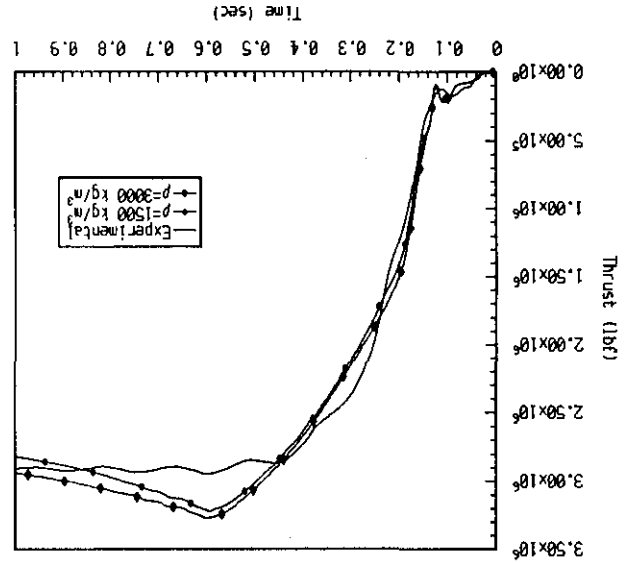


Figure 4. Thrust for $p = 1500$ and 3000 kg/m^3 and 10% aluminum oxide particles

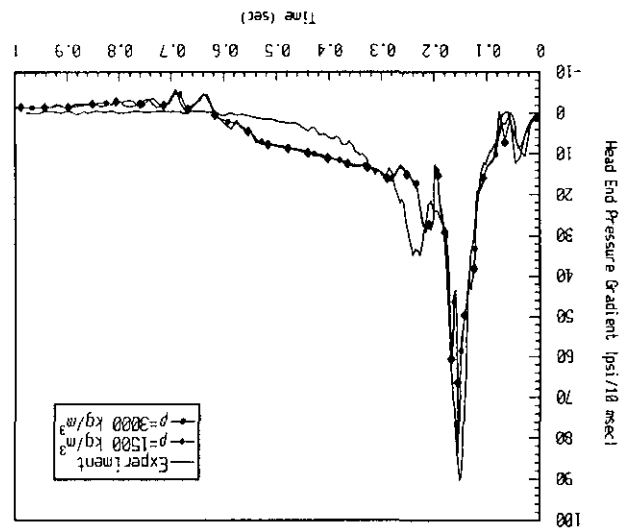


Figure 7. Effect of particles loads of 10% and 17% and $p = 3000 \text{ kg/m}^3$ on the head-end pressure gradient predictions

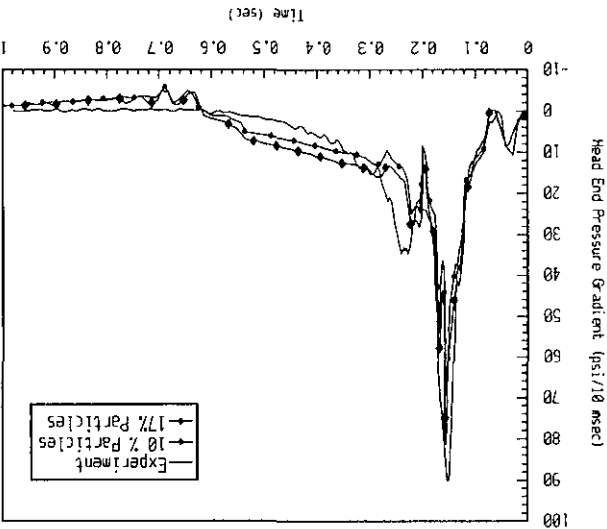
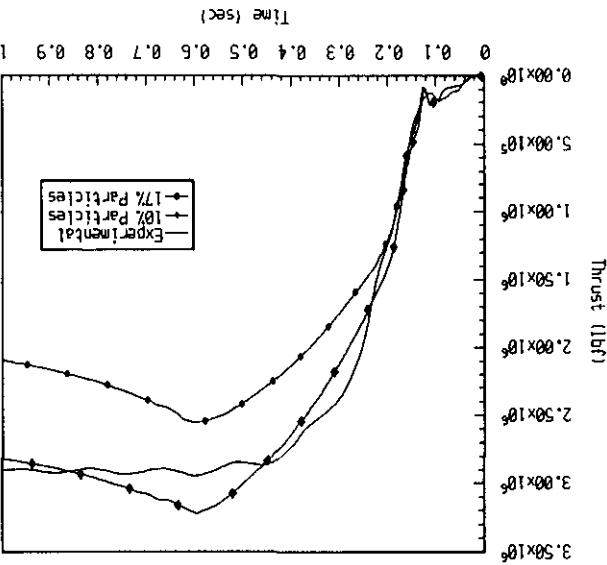


Figure 6. Effect of particle loads of 10% and 17% for $p = 3000 \text{ kg/m}^3$ on thrust predictions



The thrust predictions for the one fluid and two fluid models are shown in Figure 8. The two phase results are for a particle loading of 10% and an aluminum oxide density of 3000 kg/m^3 . The one fluid model predicts a much larger thrust build up than the measurement data. This is due to the higher flow speeds in the chamber which increase the burn rate, and the higher fluid velocities at the nozzle exit. More reasonable predictions are made with the two fluid model due to increased mixture density which slows the flow speed in the chamber. A comparison in the errors of the two models is shown in Figure 9. The two phase model shows a marked improvement in predicted thrust over the one phase model. The head-end pressure gradient results for the two models is given in Figure 10. The two fluid model shows improvement over the one fluid model during certain points in the transient process.

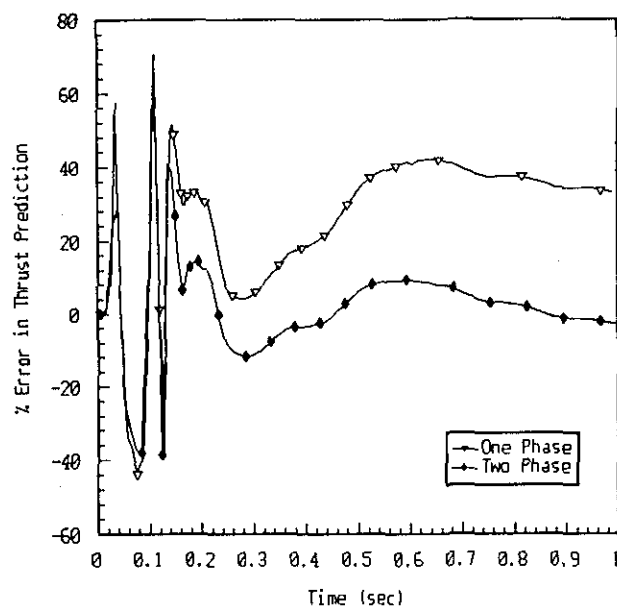


Figure 9. Error in thrust predictions for the numerical models

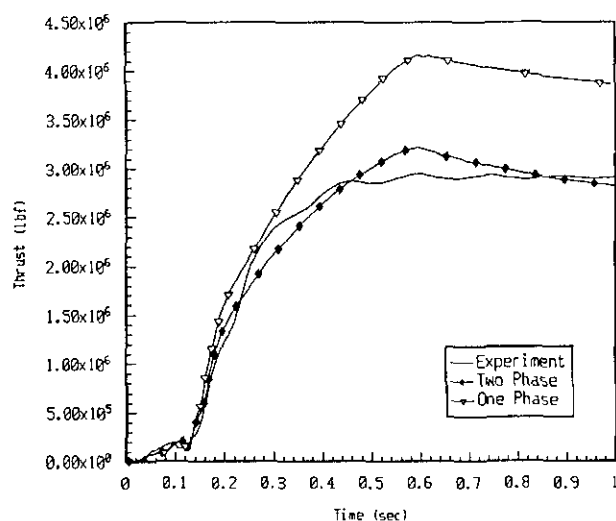


Figure 8. Comparison of overall thrust for numerical models and experimental data

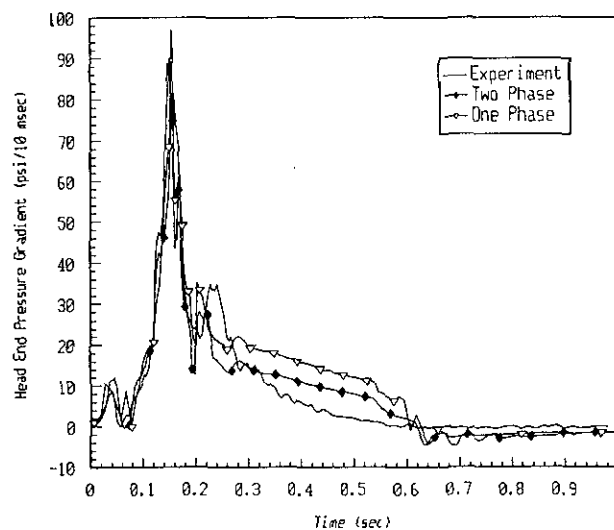


Figure 10. Comparison of head-end pressure gradient for numerical models and experimental data

It should be noted that the burning perimeter ratio, α , defined by equation 1 is arbitrarily adjusted so that numerically predicted values could match with experimental data. After many trials, $\alpha(t)$ as shown in Figure 11 was selected. All numerical results reported in this paper, both single fluid and two fluid cases, were obtained with these values of $\alpha(t)$.

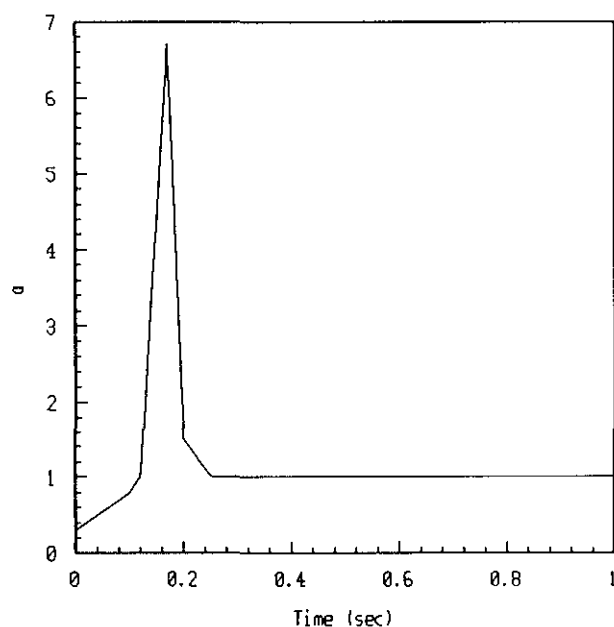


Figure 11. Variation of α with time

Conclusions

A one dimensional two fluid model has been applied to the Space Shuttle SRM ignition transient. The simulation was carried out for two different particle densities and particle loading percentages. The results show that the particle density has some effect on the predicted thrust, but it is not very large during the ignition stage. The head-end pressure gradient is found to be somewhat independent of the particle density. The particle loading shows a more pronounced effect on the predicted thrust; however, the conclusions drawn from this are limited because the flame temperature of the propellant is not modified for the presence of more particles. The two fluid model is compared to a one fluid model. The two fluid model provides an overall improvement in the solid rocket motor model. This study shows that the simulation of solid rocket motor ignition transients can be improved with the addition of two fluid effects to the numerical model.

Acknowledgements

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